

Class 00. Class Overview

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Virtual Reality

- What is virtual reality?

Virtual Reality

- What do we need to know about the VR?
 - Human
 - Machine
 - Experience

Objective of the Course

- Study physiological characteristic of human for virtual reality
- Investigate elementary feedback mechanism for virtual reality
- Understand requirements and implementation issues of virtual reality simulations
- Be able to create a virtual reality applications using HMD and SW tools

Objective of the Course

- Study physiological characteristic of human for virtual reality
 - Senses of the human
 - Experience Model
 - Perception Model (including space and time)
- Investigate elementary feedback mechanism for virtual reality
- Understand requirements and implementation issues of virtual reality simulations
- Be able to create a virtual reality applications using HMD and SW tools

Objective of the Course

- Study physiological characteristic of human for virtual reality
- Investigate elementary feedback mechanism for virtual reality
 - Interactions and Perception / Perceptual Illusion / Believability
- Understand requirements and implementation issues of virtual reality simulations
- Be able to create a virtual reality applications using HMD and SW tools

Objective of the Course

- Study physiological characteristic of human for virtual reality
- Investigate elementary feedback mechanism for virtual reality
- Understand requirements and implementation issues of virtual reality simulations
 - Hardware and Software
- Be able to create a virtual reality applications using HMD and SW tools

Objective of the Course

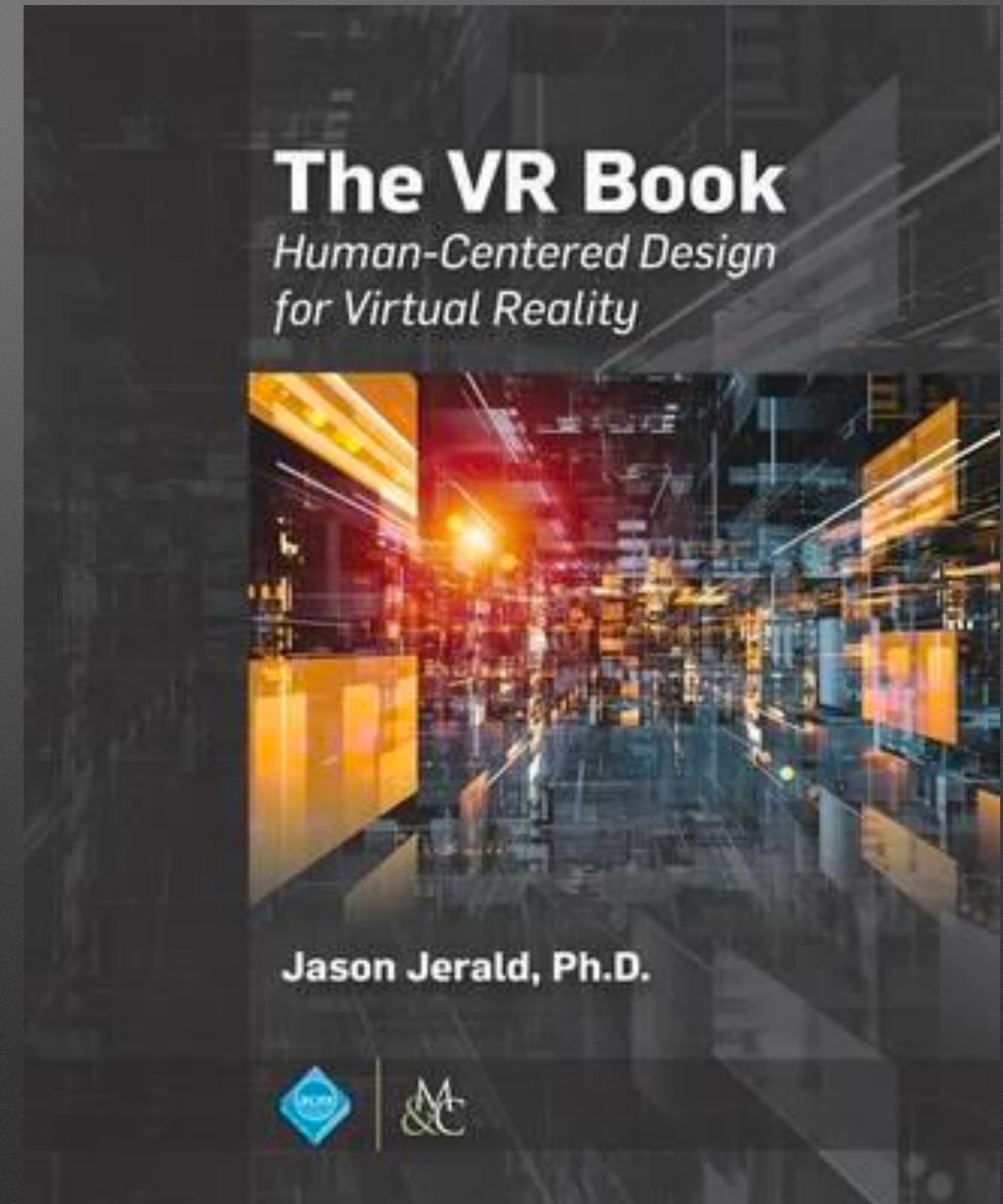
- Study physiological characteristic of human for virtual reality
- Investigate elementary feedback mechanism for virtual reality
- Understand requirements and implementation issues of virtual reality simulations
- Be able to create a virtual reality applications using HMD and SW tools
 - We will use a tool
 - Unreal 5 Engine

What you will get from the Course

- Design VR/AR Applications
- Implement VR/AR Applications
- Evaluation Experiences in VR/AR
- ...

Text Book

- eBook is also available



Term Project

- Term Projects: A team by three or four (two is possible).
- Type 1) Development of a VR application
 - Design and implement an application in VR. It maybe VR desktop application, VR game, VR tour guide, VR education, ...
- Type 2) Experiments on VR
 - Experiments on space perception, time perception, shape perception, spatial memory, etc.

Term Project Schedule

- Term Project Proposal
 - 3rd week - Need to consult a week before
- Term Project Midreport
 - 10th week
- Final Presentation
 - 15th seek

End of 1st part of 1st week